Identification and Classification of Diabetic Retinopathy Progressive Stages Using Deep Learning

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Abstract - Diabetes has emerged as a major global health challenge, with India contributing a significant share of cases, accounting for nearly one-sixth of the affected population. A major microvascular complication is diabetic retinopathy (DR), which can lead to vision loss. Traditional diagnostic procedures are both resource-intensive and slow, emphasizing the necessity of automated detection approaches. The present research investigates the application of advanced deep learning architectures—ResNet18, VGG19, MobileNetV1, and MobileNetV2—for DR detection. The objective is to determine the most suitable model that combines computational efficiency with high diagnostic accuracy and rapid training at a low cost. Our modified MobileNetV2 achieved the best results, with 88% accuracy,89% precision, 88% recall, and an F1-score of 88%.

Keywords— Deep Learning, Diabetic Retinopathy, fundus images, Data Augmentation, Convolutional neural network, Classifications and Prediction

I. INTRODUCTION

Diabetic retinopathy (DR) often develops as a complication of prolonged diabetes and is strongly associated with factors such as hypertension and inadequate blood sugar regulation. It can also contribute to renal disease, heart disease, cardiovascular mortality, and glaucoma. Although no definitive cure exists, therapeutic interventions including laser treatment, vitrectomy, and anti-VEGF drugs can significantly slow down or prevent vision impairment. [8]. A study in India (2018–2020) found that 18.8% of 42,146 participants had diabetes, with DR prevalence similar in urban (12.5%) and rural (4.0%) areas. DR was more common in diagnosed diabetics [9][10]. Diabetes affects over 536.6 million people globally, expected to rise to 783.2 million by 2045, particularly in middle-income countries. Rural populations face greater screening challenges due to age, financial constraints, and lack of guidance [11]. Many DR cases remain undetected until severe stages, highlighting the need for an automated computer-aided diagnosis (CAD) system for early detection and prevention.

Certain lesions, e.g., microaneurysms, hemorrhages, exudates, and cotton wool spots, are crucial for diagnosing and tracking the progression of DR [12]. There are two forms of DR: non-proliferative (early), and proliferative (late stage). The present work has two schemes: a binary classification (NO_DR or DR) and a multi-class classification, categorizing DR into five stages, shown in Table I.

Advances in medical technology, driven by DL, ML, and AI, have boosted healthcare efficiency, particularly computer vision-assisted diagnostics [13]. The present study compares and analyzes various CNN for classification and diagnosis of diabetic retinopathy (DR). The retinal images were labeled and used for training, testing, and validation for an exhaustive dataset. The model architecture was fine-tuned for optimal classification and its performance was calculated in terms of accuracy, precision, recall, F1-score, and support. High reliability and high accuracy are required for clinical applications [14][15]..

I. LITERATURE SURVEY

Recent developments in machine learning (ML) and deep learning (DL) have greatly enhanced the capacity to detect and classify diabetic retinopathy (DR)., the leading cause of adult blindness worldwide. This literature survey examines various methodologies, from traditional image processing to modern deep learning models, for accurate DR diagnosis and grading. Ghulam Ali et al. [1] proposed a Hybrid CNN model combining ResNet50 and InceptionV3, achieving 96.85% accuracy, 99.28% sensitivity, and 98.65% F1-score. However, its complexity and high computational demands pose challenges, especially in resource-limited settings. Their study utilized 44,119 Kaggle images, trained for 100 epochs on a Kaggle notebook with 16GB RAM and two GPUs, employing data augmentation and preprocessing techniques like histogram and intensity equalization.

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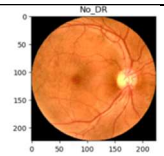
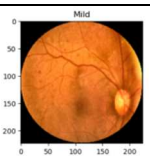
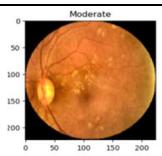
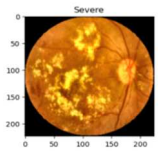
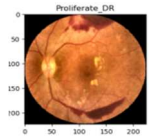
TABLE I DR PROGRESSIVE STAGES

CLASSES	0	1	2	3	4
PROGRESSIVE STAGES OF DR	NO_DR (NORMAL)	MILD NDPR	MODERATE NDPR	SEVERE NDPR	PROLIFERATE DR
DESCRIPTION	No Abnormalities	It is the early stage of DR, only microaneurysms can be observed which is very difficult to identify manually.	Any of the following lesions can be observed -microaneurysms -hemorrhages -hard exudates -cotton wool spots	Lesions together with -blockage in blood vessels of the retina -venous bleeding -intraretinal microvascular abnormalities	This is the advanced stage where any of the following can be observed -New blood vessels -pre-retinal/vitreous hemorrhage -retinal detachments
SAMPLE IMAGES					

Thangam Palaniswamy and Mahendiran Vellingiri [2] developed an IoT and deep learning-enabled diabetic retinopathy diagnosis model (IoTDL-DRD), where IoT devices collect data and transfer it to the cloud for processing. They applied Median Filtering (MF) and CLAHE for noise removal and contrast enhancement, followed by segmentation using the Mayfly Optimization-based Region Growing Technique. Features were extracted using DenseNet, and classification was performed with the HBO algorithm and LSTM model. The approach achieved 99.08% accuracy on the Messidor dataset (1,744 fundus images). However, despite high accuracy, it showed low prediction rates for Mild, Severe, and Proliferative DR classes.

Hamza Mustafa et al. [3] proposed a multistream network utilizing pretrained DL models like ResNet50 and DenseNet121 for feature extraction, PCA for dimensionality reduction, and ensemble ML classifiers (AdaBoost and Random Forest) for classification. They used multiple datasets, including Messidor-2 (with two classes: 'No Referable DME' and 'Referable DME') and EYEPAC, DDR, and APTOS (with five classes). Their model achieved accuracies of 95.58% (Messidor-2), 89.20% (EYEPAC), 89% (APTOS), and 76.81% (DDR). However, its accuracy declined when classifying five categories instead of two.

Michael Chi Seng Tang et al. [4] proposed a transfer learning-based binary classification model for neovascularization detection. They evaluated four pre-trained CNN architectures—AlexNet, GoogLeNet, ResNet18, and ResNet50—and introduced a hybrid model combining GoogLeNet and ResNet18. Fundus images were enhanced by removing the green channel and applying CLAHE. Ground truth images were manually annotated by an ophthalmologist. The model was trained and tested on 1,174 Messidor retinal images for neovascularization identification.

Sharmin Majumder and Nasser Kehtarnavaz, Deepa et al [5] proposed a multitask deep learning model for identifying all five stages of DR. Their Squeeze Excitation Densely Connected Multitasking Network integrates a squeeze-excitation network with a modified DenseNet for compression. The framework includes an MLP classifier, a regression model, and a classification model, with the pretrained Xception ImageNet model optimized for regression. Independently trained features are concatenated and fed into a multilayer perceptron for classification. The model achieved strong performance, with Weighted Kappa scores of 0.90 and 0.88 and an improved area under the ROC curve on APTOS and EyePACS datasets.

Mohamed M. Farag , Udayakumar et al. [6] proposed a DR severity grading model using DenseNet169 and Convolutional Block Attention Models. They applied auto-cropping and a 2D Gaussian filter for image enhancement and used data augmentation to address class imbalances in the APTOS dataset. A weighted training strategy based on inverse class frequency improved fairness and performance. The model achieved 97% accuracy, 97% sensitivity, 98.3% specificity, and a QWK score of 0.9455 for binary classification. It was also highly effective in severity grading, achieving 82% accuracy and a QWK of 0.888.

A Khera et al. [7] proposed a deep learning model for classifying Non-Proliferative Diabetic Retinopathy

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(NPDR) stages—mild, moderate, and severe. The model enhances NPDR detection by semantically segmenting fundus images using deep convolutional neural networks (DCNNs). It includes a lesion identification framework using Laplacian of Gaussian (LOG) and Matched Filter (MF) filters. Techniques like Transforming Curvelets for dark lesion enhancement and Bandpass Filter (BPF) optimization for bright lesions improve detection accuracy. To address class imbalance, a dense PCA-based unsupervised approach is employed. This research highlights advancements in DR detection while underscoring challenges related to model complexity, computational demands, dataset limitations, and data imbalance.

II. PROPOSED METHODOLOGY

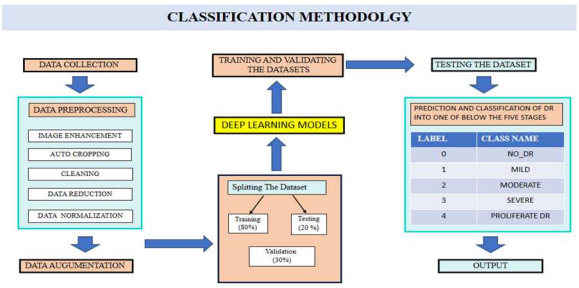

Fig.1. DR Classification Methodology

A. DATASET DEFINITION

The dataset we used in our project is taken from an open-source dataset that is available in Kaggle and was originally taken from the APTOS 2019 Blindness Detection competition training datasets. The original datasets are collected from rural areas by the technicians at Aravind Eye Hospital in India and the images are diagnosed and graded by highly trained doctors. The dataset we used consists of 3662 colored retina images and a train.csv file consisting of two columns id_code and diagnosis. All the images are readily available in their respective folders according to diagnosed classes. The distribution of the images in the dataset is as follows

TABLE II. IMAGE DISTRIBUTION WITHIN THE DATASET

Folders	NO.OF Images
NO_DR	1805
MILD	370
MODERATE	999
SEVERE	193
PROLIFERATED _DR	295

TABLE II and Fig. 2 show that the dataset is heavily imbalanced as the majority class are images with no Diabetic Retinopathyand the minority class are images with Diabetic Retinopathy. As a result, the DL models trained on this dataset were biased towards the majority class andstruggled to classify the DR progressive stages Stages.

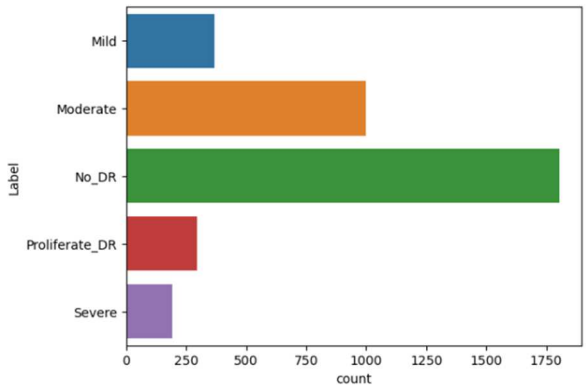

Fig. 2. Graphical Representation of the dataset distribution

B. DATA PREPROCESSING

Data preprocessing is a vital initial step in anymodel, involving the preparation of raw data to make it suitable for analysis. Its primary advantage lies in enhancing model accuracy by ensuring that only relevant and accurate information is considered, thus minimizing the impact of irrelevant or erroneous data. Additionally, data preprocessing can streamline the training process by reducing the amount of data to be processed, consequently saving time and computational resources [22][23].

1) IMAGE ENHANCEMENT

To improve image clarity, our framework applies Contrast Limited Adaptive Histogram Equalization (CLAHE) in conjunction with intensity adjustments to enhance image contrast and visibility, particularly in regions with varying illumination.

CLAHE improves upon traditional Histogram Equalization by dividing the image into small, non-overlapping tiles and applying equalization independently to each, ensuring localized contrast enhancement while preventing over-amplification of noise in low-contrast areas. Additionally, intensity adjustment using the `convertScaleAbs()` function fine-tunes image brightness and contrast by applying scaling factors (alpha and beta) to pixel values. This combination results in clearer, more detailed images, as illustrated in Fig. 3.

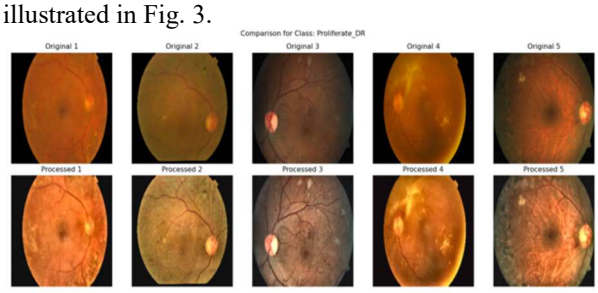

Fig. 3. The fundus images before and after applying the image enhancement technique

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2) CROPPING

Image cropping is carried out in two stages: an automated circular crop followed by fine auto-cropping for precision. Initially, the image is transformed to grayscale and smoothed with Gaussian blur to minimize noise, followed by Otsu's thresholding to create a binary image. The process detects contours, isolating the largest one as the focal area. A circular mask, based on this contour's bounding box, is applied to the image, isolating a circular region through a bitwise AND operation. Subsequently, this region is refined using auto-cropping to eliminate any residual background, which involves additional grayscale conversion, threshold application, and contour detection to precisely crop the image around the primary subject, resulting in a cleanly cropped final image.

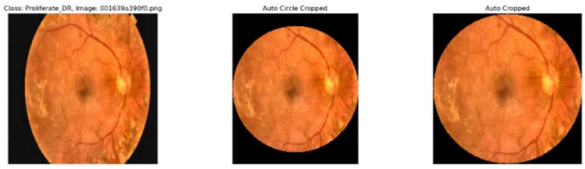

Fig. 4. A sample image from Proliferate_DR before and after applying Auto Circle Crop and Auto Crop.

3) CLEANING

After auto cropping, manual cleaning is done on the dataset because the automated process failed to perfectly isolate a few images from the region of interest (ROI) from the background and inaccurately cropped parts of the subject. This inconsistency can be due to various factors such as uneven lighting, similar textures between the subject and background, or irregular shapes that don't conform well to the algorithm's expectations [24][25][26]. The goal of manual cleaning is to ensure that the final set of images is as close to the desired outcome as possible, maintaining consistency and quality.

4) DATA REDUCTION

One of the methods employed for balancing the dataset was random under sampling, which reduces the size of the majority class. It entails randomly deleting instances from the majority class (No_DR and Moderate) in an unbalanced dataset in order to reduce its size and eventually attain a more balanced distribution across the classes. This strategy seeks to address the issue of class disparity, Random under sampling contributes to the creation of a more balanced dataset by lowering the number of cases in the majority class. This can enhance model performance, particularly for classification problems.

5) NORMALIZATION

Normalization refers to rescaling image pixel values into a fixed range to standardize input data for training. so that they fall within a consistent range, often between 0 and 1 or -1 to 1. In our models, the `rescale=1./255` parameter transforms the pixel values to lie within a 0 to 1 scale. Such a step is critical for facilitating effective learning by the model, as it ensures uniformity in the scale of input data, which is beneficial for the optimization algorithms used during the model's training phase.

C. DATA AUGMENTATION

To mitigate class imbalance, data augmentation strategies were applied like rotation, width, and height shifting, shearing, zooming, and horizontal flipping to oversample the image on minority classes such as Mild, severe, and Proliferate DR respectively the below Fig. 5 represents the distribution of the balanced dataset where are there 400 images on the Mild, Moderate, Severe and Proliferate DR classes and 1600 images on No_DR class. A total of 3200 images were used.

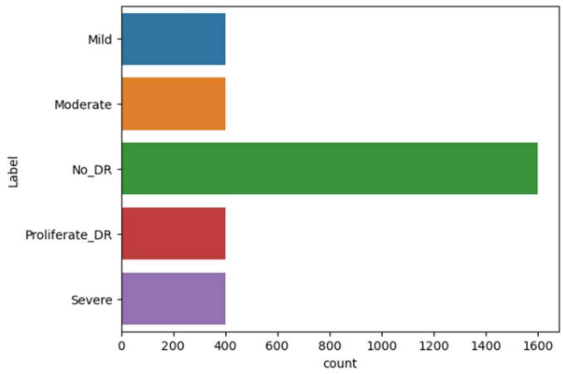

Fig. 5. Graphical Representation of balanced dataset.

By balancing the dataset, a more diverse set of examples provides for the model to learn from, enhancing its ability to generalize to unseen data. This prevents the model from being biased towards the majority class during training. Additionally, it introduces variability into the training data, reducing overfitting and encouraging the model to learn features invariant to transformations. Augmentation acts as regularization by adding noise to the training process, preventing the model from memorizing the data.

D. MODEL TRAINING

For DR classification, four different convolutional neural network models were implemented— ResNet18, VGG19, MobileNetV1, and MobileNetV2—were utilized, each tailored with a consistent training approach to ensure comparability of results. These models were trained over 100 epochs using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy loss function, accommodating a batch size of 32. The dataset was partitioned into an 80% training set, with 20% reserved for testing and an additional 30% of the training data allocated for validation purposes. To enhance the model's robustness and address the potential for overfitting, data

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augmentation techniques were applied exclusively to the training dataset. Moreover, each model's input images were resized to a target size specific to the architecture's requirements. The training process was conducted in Jupyter Notebooks within the Anaconda Navigator environment, employing key techniques to improve model performance and prevent overfitting: EarlyStopping was configured to halt training if the validation loss ceased to improve for 10 epochs, ModelCheckpoint saved only the best-performing model weights, and ReduceLROnPlateau reduced the learning rate by a factor of 0.1 if there was no improvement in validation loss after 5 epochs, with a minimum learning rate set at 1e-6. These strategies collectively aimed to enhance the accuracy and efficiency of diagnosing diabetic retinopathy through deep learning models.

1. ResNet18

ResNet-18 tackles complex image data with surprising efficiency. This 18-layer convolutional neural network utilizes residual connections, a clever shortcut that allows information to flow directly through the network, avoiding the vanishing gradient problem that plagues deep architectures. By combining convolutional layers for feature extraction with max-pooling for focused analysis, ResNet-18 progressively captures intricate visual details. These residual connections not only boost accuracy but also make the network memory-efficient, making it a powerful tool for various computer vision tasks.

2. VGG 19

The our proposed VGG19 architecture the model is without the fully connected layer and trained for 20 epochs. Later These pre-trained layers are t frozen to prevent their weights from being updated during training. Subsequently, two new fully connected layers with ReLU activation functions are added, followed by dropout layers with a dropout rate of 0.5 to prevent overfitting. The final output layer consists of Dense neurons equal to the number of classes, with softmax activation for multi-class classification. The model is compiled with the option to fine-tune the last few layers by making them trainable.

3. MobileNetV1 and MobileNetV2

MobileNet is a lightweight CNN architecture optimized for mobile and embedded vision applications. It uses depthwise separable convolutions to reduce parameters and computational complexity while maintaining high accuracy. In our proposed system, pre-trained MobileNet models were used without their top classification layers, allowing for feature extraction from input images. Custom classification layers were added to adapt the model for diabetic retinopathy classification. The model was compiled using the Adam optimizer and categorical cross-entropy loss function, preparing it for training on the dataset..

III. RESULT AND DISCUSSIONS

The evaluation results in Table III highlight MobileNetV2's superior performance over ResNet18, VGG19, and MobileNetV1 in DR classification. Achieving 88% accuracy, MobileNetV2 demonstrates its effectiveness in identifying DR cases. Its precision of 89% indicates strong true positive identification, while a recall rate of 88% shows its ability to capture actual positive cases. The F1 score of 88% balances precision and recall, reinforcing its robustness in minimizing false positives and false negatives. These metrics establish MobileNetV2 as a highly effective model for DR detection, making it a promising choice for medical image analysis.

TABLE III RESULT OBTAINED FROM THE OPTIMIZED MODELS

MODEL	ResNet18	VGG 19	MobileNetV1	MobileNetV2
ACCURACY	74%	83%	79%	88%
PRECISION	74%	84%	81%	89%
RECALL	72%	83%	79%	88%
F1-SCORE	74%	83%	79%	88%

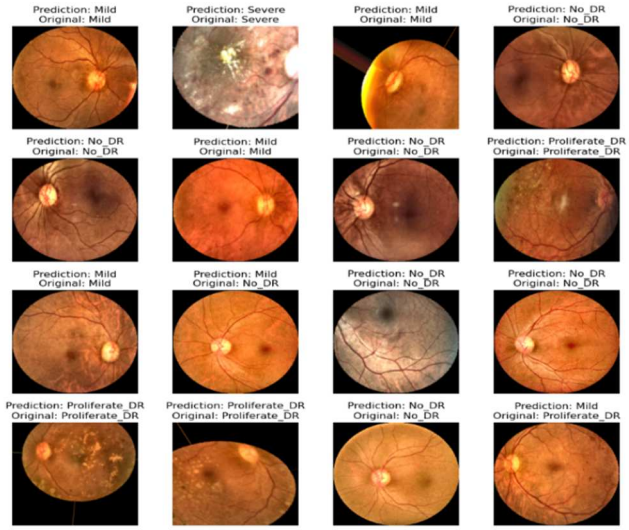

Fig.6. Prediction Result by theMobileNetV2 model

	precision	recall	f1-score	support
Mild	0.78	0.82	0.80	87
Moderate	0.66	0.79	0.72	77
No_DR	0.97	0.99	0.98	321
Proliferate_DR	0.85	0.74	0.79	77
Severe	0.92	0.73	0.81	78
accuracy			0.88	640
macro avg	0.84	0.81	0.82	640
weighted avg	0.89	0.88	0.88	640

Fig.7 DR Classification Report of MobiNetV2 model

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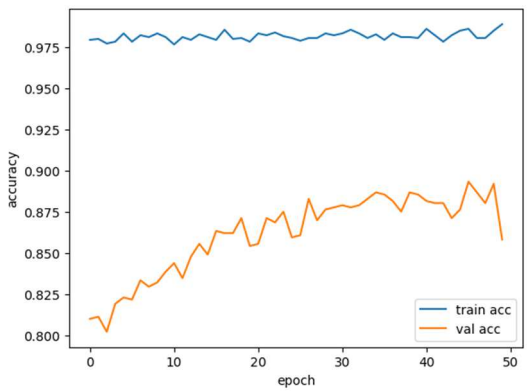

Fig.8. MobileNetV2 Accuracy Graph

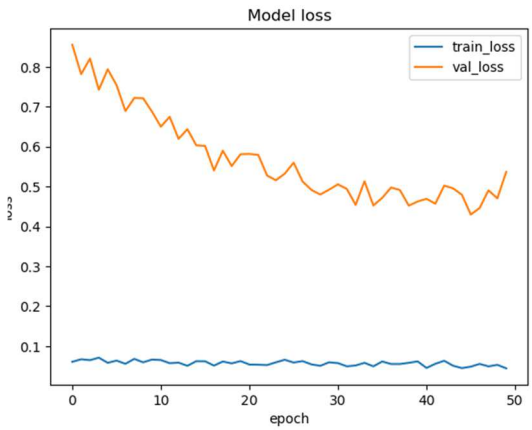

Fig.9. MobileNetV2 Loss Graph

V. CONCLUSION

Early screening for DR remains difficult in rural regions because of restricted healthcare availability, low awareness, inadequate infrastructure, and geographical barriers. However, computer-aided diagnosis (CAD) systems using ML and DL offer a promising solution by enabling remote screening through mobile devices, automating retinal image analysis with low-cost cameras, and aiding early detection. This study proposed an optimal deep learning model for DR detection, prioritizing accuracy, low computational complexity, and cost efficiency. Among the tested models, MobileNetV2 achieved the highest accuracy (88%) with the best classification and prediction rates. Its lightweight architecture ensured fast predictions (≤1s/step) while being cost-effective. These improvements were achieved through data augmentation and preprocessing techniques that enhanced image quality. Future work will explore additional deep learning and hybrid models to further improve classification accuracy and achieve more precise results.

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